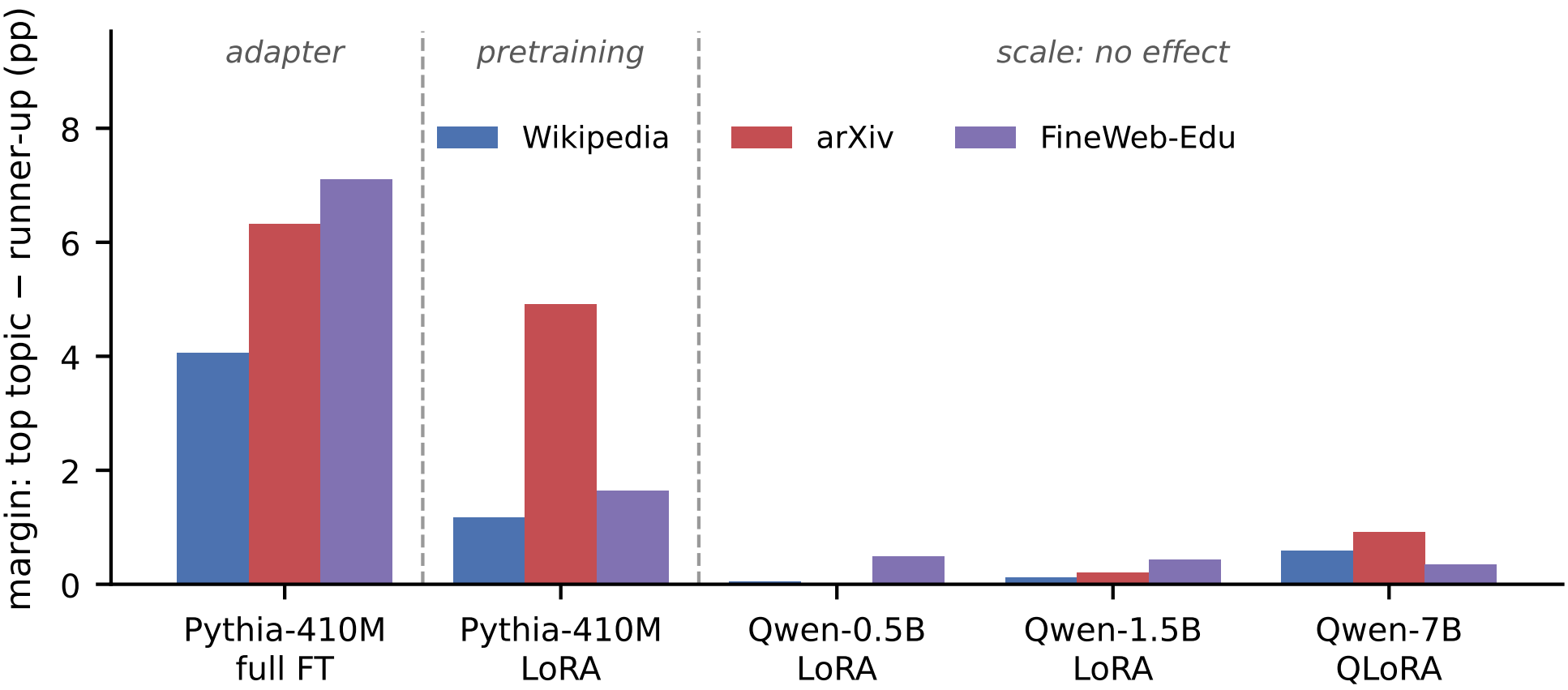
adapter
pretraining
scale: no effect
Wikipedia
arXiv
FineWeb-Edu
margin: top topic − runner-up (pp)
0
2
4
6
8
Pythia-410M
full FT
Pythia-410M
LoRA
Qwen-0.5B
LoRA
Qwen-1.5B
LoRA
Qwen-7B
QLoRA